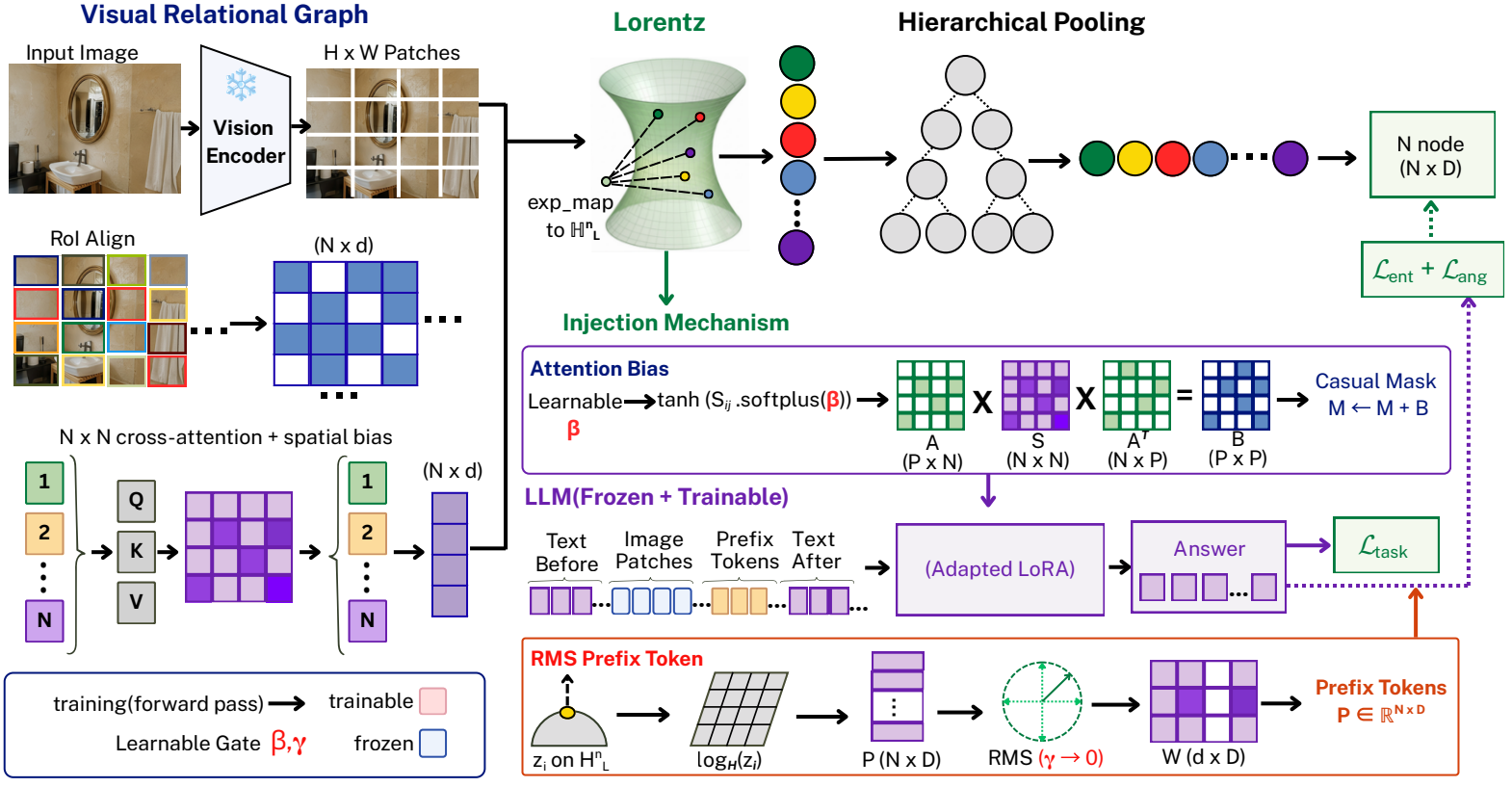

Visual Relational Graph
Input Image
Vision Encoder
H x W Patches
RoI Align
(N x d)
N x N cross-attention + spatial bias
1
2
N
Q
K
V
(N x d)
training(forward pass)
trainable
Learnable Gate β,γ
frozen
Lorentz
exp_map to $\mathbb{H}^n_L$
Injection Mechanism
Hierarchical Pooling
N node (N x D)
$\mathcal{L}_{ent} + \mathcal{L}_{ang}$
Attention Bias
Learnable → tanh (S_{ij} .softplus(β))
β
A (P x N)
X
S (N x N)
X
A^T (N x P)
=
B (P x P)
Casual Mask
M ← M + B
LLM(Frozen + Trainable)
Text Before
Image Patches
Prefix Tokens
Text After
(Adapted LoRA)
Answer
$\mathcal{L}_{task}$
RMS Prefix Token
z_i on H^n_L
$\log_H(z_i)$
P (N x D)
RMS (γ → 0)
W (d x D)
Prefix Tokens
$P \in \mathbb{R}^{N \times D}$